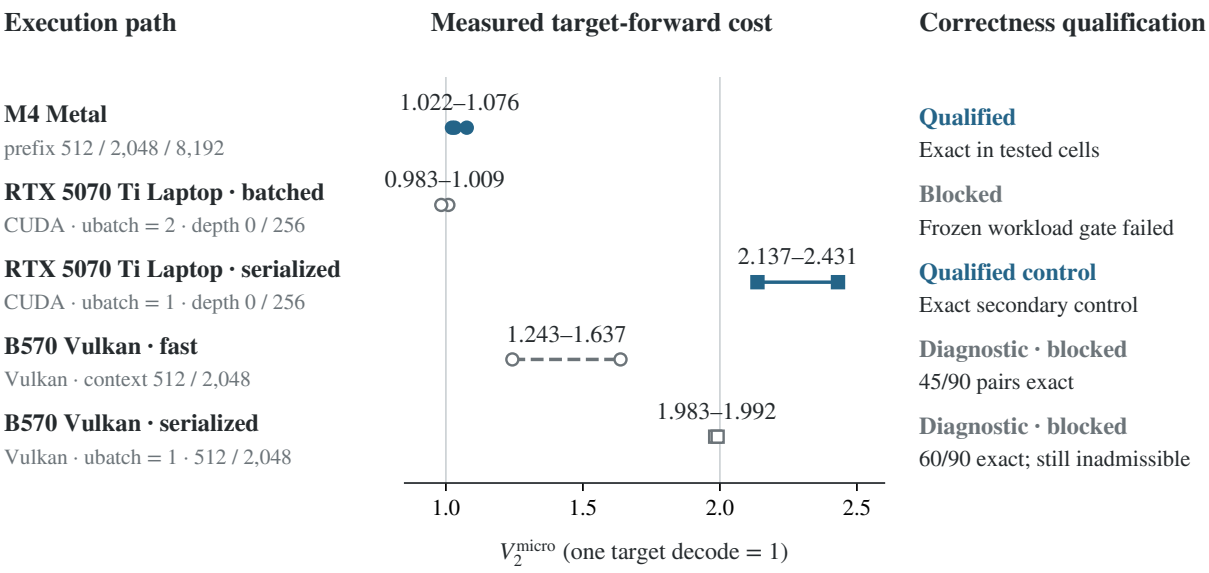
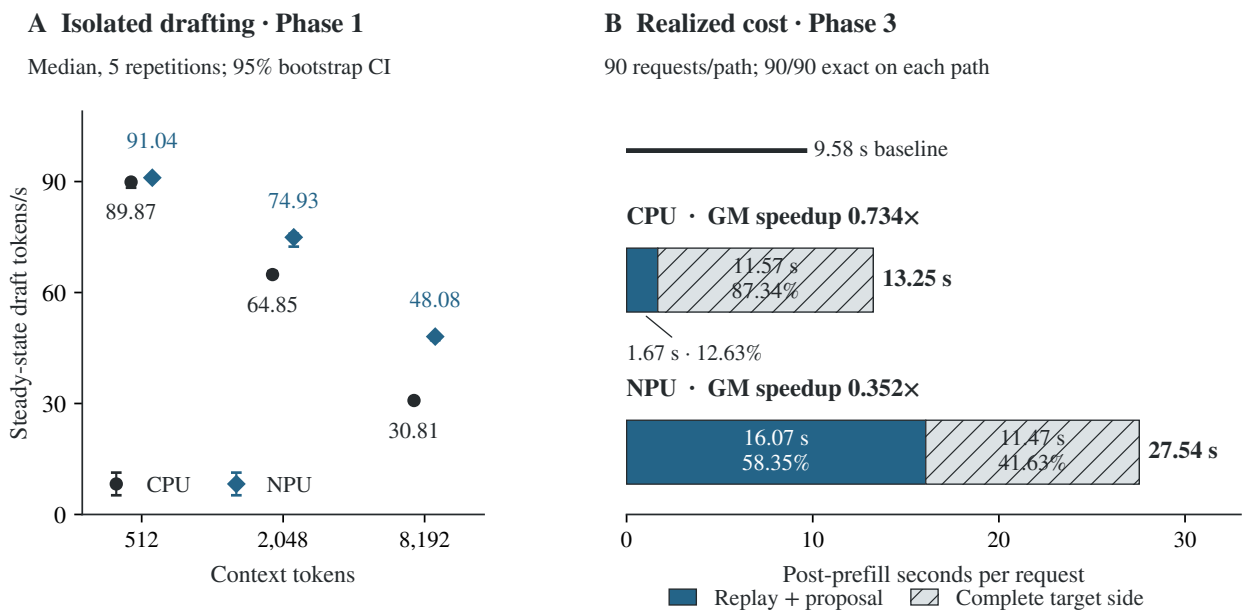


Filled = qualified; open = blocked/diagnostic. Circles = batched/fast; squares = serialized.
Ranges span tested contexts (not CIs); contexts are not fully matched. $V_2^{micro} \neq R_{target}$.

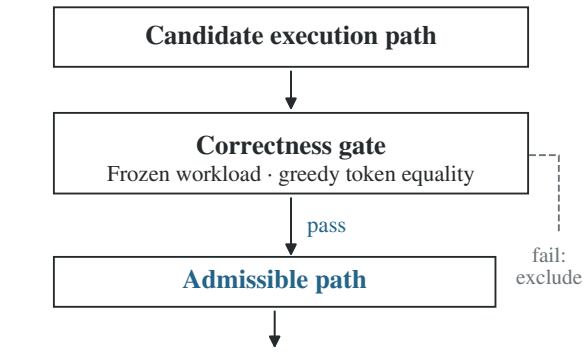


Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.
Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.
Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.



Measure realized work per round

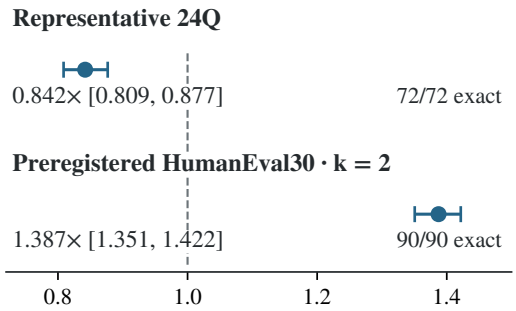
D_{real}	Draft through its actual state interface
R_{target}	Complete target-side round work
X	Residual cost, without double counting
C	All output tokens committed per round

$$S = C / (D_{\text{real}} + R_{\text{target}} + X)$$

Costs normalized by T_b; one consistent boundary.

Separate proxy: V_k^{micro} is not complete R_{target}.

M4 Metal: two frozen workload studies



95% prompt-bootstrap CIs; separate workload studies.
Exact = token equality, not functional HumanEval pass@k.